

HYDROGENATED NBR SYNTHESIS AND PROCESSING PATENT LANDSCAPE

1. ABSTRACT

This report synthesizes nine patents relevant to hydrogenated nitrile rubber (HNBR), spanning direct hydrogenation of NBR, simultaneous hydrogenation/metathesis routes, catalyst-residue control, and downstream compounding or application-driven processing. The target material is defined as nitrile-butadiene rubber with reduced residual unsaturation, typically retaining the nitrile-containing backbone while converting a substantial fraction of butadiene-derived double bonds to saturated linkages to improve heat, oil, and ozone resistance. Across the selected patents, the dominant synthesis strategies are: (1) homogeneous noble-metal hydrogenation of NBR in solution or cement, often using ruthenium, palladium, or rhodium catalysts; (2) one-step hydrogenation plus metathesis to lower molecular weight and narrow polydispersity; and (3) post-hydrogenation processing into blends, tires, or bonded flexible pipes where HNBR performance is leveraged.

The landscape shows two major process directions. First, metathesis-enabled depolymerization/hydrogenation patents (US7919563B2, US7772328B2, EP2473280B1, US9150669B2) aim to reduce molecular weight and Mooney viscosity while maintaining the HNBR microstructure. These disclosures emphasize ruthenium/osmium carbene catalysts, co-olefins such as ethylene or 1-hexene, and reaction conditions around 80 °C and elevated pressure. Second, conventional hydrogenation patents (EP3827029A1) focus on catalyst selection, catalyst loading, and especially catalyst removal/recovery to avoid discoloration and aging penalties; they specify Ru, Pd, and Rh loadings and broad hydrogenation windows. The remaining patents are application/process patents that incorporate HNBR into blends or bonded structures and disclose practical compounding parameters such as peroxide type, scorch time, acrylonitrile content, residual double bond content, and cure temperature.

Overall, the evidence supports HNBR synthesis routes that combine controlled hydrogenation with either metathesis-assisted molecular-weight reduction or careful catalyst management. The most technically relevant parameters are moderate hydrogenation temperatures, sufficient hydrogen pressure, low catalyst loadings, and post-reaction purification or catalyst recovery to preserve aging performance and color stability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20130303700A1
- Patent Title: Blends of partially hydrogenated nitrile rubber and silicone rubber, ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses producing blends of partly hydrogenated nitrile rubber and vinyl-containing silicone rubber by mixing the two rubbers in the presence of a metathesis catalyst. The process is carried out essentially solvent-free and uses a ruthenium carbene catalyst during kneading/mixing to promote interpolymer metathesis and improve blend morphology. - Partly hydrogenated nitrile rubber acrylonitrile content: 34% by weight — Example 1 - Partly hydrogenated nitrile rubber residual double bond content: 5.5% — Example 1 - Partly hydrogenated nitrile rubber Mooney viscosity ML(1+4) at 100°C: 68 ME — Example 1 - Silicone rubber vinyl group content: 5 mol % — Example 1 - Silicone rubber number-average molecular weight Mn: 138000 g/mol — Example 1 - Silicone rubber weight-average molecular weight Mw: 304000 g/mol — Example 1 - Catalyst amount: 0.1 parts by weight — Example 1 (inventive blends 1.3, 1.5, 1.7) - Brabender capacity: 350 ml — Example 1 - Fill level: 70% — Example 1 - Rotor speed: 50 min⁻¹ — Example 1 - Start temperature: 35°C — Example 1 - Initial kneading time: 2 min — Example 1 - Addition window for silicone rubber and catalyst: within 5 min — Example 1 - Subsequent kneading time: 60 min — Example 1 - Internal temperature: 60°C — Example 1 - Room-temperature storage between mixing stages: 24 h — Example 1

Relevant Experimental Evidence - Example 1: HNBR/silicone blends were prepared with and without metathesis catalyst, showing catalyst-assisted mixing routes. - Example 1: The HNBR feedstock was a partly hydrogenated nitrile rubber with 34% acrylonitrile and 5.5% residual double bonds. - Example 1: Mixing at 35°C followed by 60°C kneading demonstrated a practical solvent-free compounding route. - Example 1: Multi-stage catalyst addition was used in some inventive blends to vary blend development.

Technical Relevance Although not a synthesis of HNBR itself, this patent is relevant because it uses partly hydrogenated nitrile rubber as the key elastomer and shows how metathesis catalysts can be integrated into HNBR processing. It provides practical compounding conditions and confirms the importance of residual unsaturation control in HNBR-based materials.

Patent 2

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent prepares low molecular weight HNBR by subjecting nitrile rubber to a metathesis reaction in the presence of ruthenium or osmium catalysts, optionally with a co-olefin and solvent. The process is designed to cleave polymer chains at unsaturation sites, lowering molecular weight and Mooney viscosity while retaining the hydrogenated nitrile rubber structure. - Rubber dissolved amount: 200 g — Example 1 - MCB amount: 1133 g — Example 1 - Solid concentration: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber dissolved amount: 75 g — Example 3 - MCB amount: 1175 g — Example 3 - Solid concentration: 6 wt.-% — Example 3 - 1-hexene amount: 150 g — Example 4 - Nitrogen degassing: 3 times — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — typical product description - Mw: 54 kg/mol — typical product description - MWD: 2.0 — typical product description - Starting polymer Mn: 85 kg/mol — typical product description - Starting polymer Mw: 296 kg/mol — typical product description - Starting polymer MWD: 3.4 — typical product description - Experiment 1 Mn: 73000 — Table 2 - Experiment 1 Mw: 189000 — Table 2 - Experiment 1 PDI: 2.59 — Table 2 - Experiment 1 Mooney viscosity: 43 — Table 2 - Experiment 2 Mn: 60000 — Table 2 - Experiment 2 Mw: 136000 — Table 2 - Experiment 2 PDI: 2.27 — Table 2 - Experiment 2 Mooney viscosity: 28 — Table 2 - Experiment 3 Mn: 31000 — Table 2 - Experiment 3 Mw: 59000 — Table 2 - Experiment 3 PDI: 1.90 — Table 2 - Experiment 3 Mooney viscosity: 3 — Table 2 - Experiment 4 Mn: 55000 — Table 2 - Experiment 4 Mw: 111000 — Table 2 - Experiment 4 PDI: 2.02 — Table 2 - Experiment 4 Mooney viscosity: 31 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis in 15 wt.-% cement at 80°C reduced molecular weight and viscosity. - Example 2: Repeated ethylene metathesis under similar conditions provided a second low-MW HNBR sample. - Example 3: Lower-solids cement (6 wt.-%) also produced low-MW HNBR with narrower distribution. - Example 4: 1-hexene was used as co-olefin, demonstrating chain-transfer control with a different olefin. - Example 4: The product retained low Mooney viscosity and reduced PDI relative to the starting polymer.

Technical Relevance This is one of the core HNBR synthesis patents because it directly converts nitrile rubber into low-MW HNBR using metathesis chemistry. It establishes catalyst loading, temperature, pressure, and co-olefin effects as key levers for controlling processability.

Patent 3

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is closely related to US7919563B2 and discloses the same low molecular weight HNBR preparation concept via metathesis of nitrile rubber in the presence of ruthenium or osmium catalysts. The process uses co-olefins such as ethylene or 1-hexene to drive chain scission and reduce molecular weight and polydispersity. - Rubber dissolved amount: 200 g — Example 1 - MCB amount: 1133 g — Example 1 - Solid concentration: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber dissolved amount: 75 g — Example 3 - MCB amount: 1175 g — Example 3 - Solid concentration: 6 wt.-% — Example 3 - 1-hexene amount: 150 g — Example 4 - Nitrogen degassing: 3 times — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — typical product description - Mw: 54 kg/mol — typical product description - MWD: 2.0 — typical product description - Starting polymer Mn: 85 kg/mol — typical product description - Starting polymer Mw: 296 kg/mol — typical product description - Starting polymer MWD: 3.4 — typical product description - Experiment 1 Mn: 73000 — Table 2 - Experiment 1 Mw: 189000 — Table 2 - Experiment 1 PDI: 2.59 — Table 2 - Experiment 1 Mooney viscosity: 43 — Table 2 - Experiment 2 Mn: 60000 — Table 2 - Experiment 2 Mw: 136000 — Table 2 - Experiment 2 PDI: 2.27 — Table 2 - Experiment 2 Mooney viscosity: 28 — Table 2 - Experiment 3 Mn: 31000 — Table 2 - Experiment 3 Mw: 59000 — Table 2 - Experiment 3 PDI: 1.90 — Table 2 - Experiment 3 Mooney viscosity: 3 — Table 2 - Experiment 4 Mn: 55000 — Table 2 - Experiment 4 Mw: 111000 — Table 2 - Experiment 4 PDI: 2.02 — Table 2 - Experiment 4 Mooney viscosity: 31 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene metathesis at 80°C and 500 psi-class pressure reduced molecular weight. - Example 2: Similar conditions reproduced the low-MW HNBR outcome. - Example 3: Lower solids loading still gave strong molecular-weight reduction. - Example 4: 1-hexene co-olefin demonstrated alternative chain-transfer control. - Example 4: The data show a strong drop in Mooney viscosity and PDI versus the starting polymer.

Technical Relevance This patent is technically important because it confirms the metathesis-based route to low-MW HNBR and provides a second disclosure of the same process family. It reinforces the role of co-olefins, catalyst loading, and temperature in tailoring HNBR processability.

Patent 4

- Patent Number: US10759885B2
- Patent Title: Hydrogenated branched conjugated diene copolymer, rubber composition and ...
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparing hydrogenated branched conjugated diene copolymers by first copolymerizing branched dienes such as myrcene or farnesene with vinyl monomers, then hydrogenating the resulting copolymer. The hydrogenated branched copolymer is used in tire rubber compositions to improve processability while maintaining performance. - Polymerization vessel volume: 3 liters — Preparation Example 1 - Hexane amount: 2000 ml — Preparation Example 1 - Butadiene amount: 110 g — Preparation Example 1 - Styrene amount: 90 g — Preparation Example 1 - TMEDA amount: 0.22 mmol — Preparation Example 1 - n-BuLi amount: 35 mmol — Preparation Example 1 - Polymerization time: 5 hours — Preparation Example 1 - Polymerization temperature: 50°C — Preparation Example 1 - Isopropanol/hexane quench: 1.15 ml of 1M solution — Preparation Example 1 - Hydrogenation catalyst: 10 g of 10% palladium carbon — Preparation Example 2 - Hydrogenation pressure: 5.0 kgf/cm² — Preparation Example 2 - Hydrogenation temperature: 80°C — Preparation Example 2 - Degree of hydrogenation: 50% — Preparation Example 2 - Polymerization vessel volume: 1 liter — Preparation Example 3 - Hexane amount: 500 ml — Preparation Example 3 - THF amount: 46 g — Preparation Example 3 - n-BuLi amount: 45 mmol — Preparation Example 3 - Isoprene amount: 150 g — Preparation Example 3 - Styrene amount: 125 g — Preparation Example 3 - Dropwise addition time: 2 hours — Preparation Example 3 - Isopropanol/hexane quench: 20 ml of 2M solution — Preparation Example 3 - Polymerization vessel volume: 1 liter — Preparation Example 5 - Hexane amount: 500 ml — Preparation Example 5 - THF amount: 46 g — Preparation Example 5 - n-BuLi amount: 40 mmol — Preparation Example 5 - Myrcene amount: 150 g — Preparation Example 5 - Styrene amount: 125 g — Preparation Example 5 - Isopropanol/hexane quench: 10 ml of 2M solution — Preparation Example 5 - Polymerization vessel volume: 1 liter — Preparation Example 7 - Hexane amount: 500 ml — Preparation Example 7 - THF amount: 46 g — Preparation Example 7 - n-BuLi amount: 40 mmol — Preparation Example 7 - Butadiene amount: 95 g — Preparation Example 7 - Myrcene amount: 55 g — Preparation Example 7 - Styrene amount: 125 g — Preparation Example 7 - Isopropanol/hexane quench: 10 ml of 2M solution — Preparation Example 7 - Butadiene amount replaced by isoprene: 95 g — Preparation Example 9 - Hydrogenation catalyst: 10 g of 10% palladium carbon — Preparation Example 12 - Hydrogenation pressure: 5.0 kgf/cm² — Preparation Example 12 - Hydrogenation temperature: 80°C — Preparation Example 12 - Degree of hydrogenation: 50% — Preparation Example 12 - Polymerization vessel volume: 1 liter — Preparation Example 13 - Hexane amount: 500 ml — Preparation Example 13 - THF amount: 46 g — Preparation Example 13 - n-BuLi amount: 45 mmol — Preparation Example 13 - Isoprene amount: 150 g — Preparation Example 13 - Styrene amount: 125 g — Preparation Example 13 - Dropwise addition time: 2 hours — Preparation Example 13 - Isopropanol/hexane quench: 20 ml of 2M solution — Preparation Example 13 - Farnesene amount: 150 g — Preparation Example 15 - Styrene amount: 125 g — Preparation Example 15 - n-BuLi amount: 40 mmol — Preparation

Example 15 - Isopropanol/hexane quench: 10 ml of 2M solution — Preparation Example 15 - Farnesene amount: 55 g — Preparation Example 17 - Butadiene amount: 95 g — Preparation Example 17 - Isoprene used instead of butadiene: 95 g — Preparation Example 19

Relevant Experimental Evidence - Preparation Example 1: A branched diene/styrene copolymer was synthesized by n-BuLi initiation at 50°C and nearly 100% conversion. - Preparation Example 2: The copolymer was hydrogenated over Pd/C at 80°C and 5.0 kgf/cm² to 50% hydrogenation. - Preparation Example 3: Isoprene/styrene copolymerization provided a second precursor architecture for hydrogenation. - Preparation Example 7: A ternary butadiene/myrcene/styrene copolymer was prepared, showing branched-diene incorporation. - Preparation Example 15: Farnesene/styrene copolymerization demonstrated another branched-diene route for HNBR precursors.

Technical Relevance This patent is relevant because it defines a branched-diene precursor route followed by hydrogenation, expanding the HNBR family beyond conventional butadiene-acrylonitrile feedstocks. It is useful for understanding how precursor architecture and partial hydrogenation can tune processability and tire performance.

Patent 5

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen, optionally a co-olefin, and a ruthenium or osmium catalyst. The process is intended to produce HNBR with lower molecular weight, narrower molecular weight distribution, and low residual double bond content in a single step. - Hydrogenated nitrile rubber molecular weight Mw: 20,000 to 250,000 — invention summary - Hydrogenated nitrile rubber Mooney viscosity ML 1+4 at 100°C: 1 to 50 — invention summary - Hydrogenated nitrile rubber MWD/PDI: less than 2.8 — invention summary - Nitrile rubber Mooney viscosity ML 1+4 at 100°C: 24 to 70 — description - Preferred nitrile rubber Mooney viscosity ML 1+4 at 100°C: 28 to 40 — description - Hydrogenated nitrile rubber catalyst amount: 0.057 phr — Example 1, HNBR 1 - Hydrogenated nitrile rubber catalyst amount: 0.080 phr — Example 1, HNBR 2 - Hydrogenated nitrile rubber catalyst amount: 0.06 phr — Example 1, HNBR 3 - Hydrogenated nitrile rubber catalyst amount: 0.055 phr — Example 1, HNBR 4 - Metal content: 65.6 ppm — Example 1, HNBR 1 - Metal content: 91.3 ppm — Example 1, HNBR 2 - Metal content: 65.6 ppm — Example 1, HNBR 3 - Metal content: 65.6 ppm — Example 1, HNBR 4 - Reaction time: 0 min — Table 2 - Reaction time: 60 min — Table 2 - Reaction time: 120 min —

Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 100% at 0 min — Table 2 - RDB: 2.0% at 60 min for HNBR 1 — Table 2 - RDB: 0.7% at 60 min for HNBR 2 — Table 2 - RDB: 15.4% at 60 min for HNBR 3 — Table 2 - RDB: 2.9% at 60 min for HNBR 4 — Table 2 - RDB: 0.4% at 120 min for HNBR 1 — Table 2 - RDB: 0.5% at 120 min for HNBR 2 — Table 2 - RDB: 3.1% at 120 min for HNBR 3 — Table 2 - RDB: 0.6% at 120 min for HNBR 4 — Table 2 - RDB: 0.3% at 180 min for HNBR 1 — Table 2 - RDB: 0.3% at 180 min for HNBR 2 — Table 2 - RDB: 1.3% at 180 min for HNBR 3 — Table 2 - RDB: 0.3% at 180 min for HNBR 4 — Table 2 - RDB: 0.6% at 240 min for HNBR 3 — Table 2 - Mw: 191541 g/mol — Table 3, HNBR 1 - Mn: 68643 g/mol — Table 3, HNBR 1 - PDI: 2.79 — Table 3, HNBR 1 - Mooney viscosity: 38.2 — Table 3, HNBR 1 - Mw: 135772 g/mol — Table 3, HNBR 2 - Mn: 58259 g/mol — Table 3, HNBR 2 - PDI: 2.33 — Table 3, HNBR 2 - Mooney viscosity: 20.2 — Table 3, HNBR 2 - Mw: 195304 g/mol — Table 3, HNBR 3 - Mn: 64130 g/mol — Table 3, HNBR 3 - PDI: 3.05 — Table 3, HNBR 3 - Mooney viscosity: 69.3 — Table 3, HNBR 3 - Mw: 144212 g/mol — Table 3, HNBR 4 - Mn: 59851 g/mol — Table 3, HNBR 4 - PDI: 2.45 — Table 3, HNBR 4 - Mooney viscosity: 32.6 — Table 3, HNBR 4

Relevant Experimental Evidence - Example 1: Four catalyst systems were compared for simultaneous hydrogenation/metathesis of NBR. - Example 1: Grubbs III gave faster hydrogenation than Grubbs II or Wilkinson's catalyst at equal metal content. - Example 1: HNBR 1 and HNBR 2 showed lower Mooney viscosity than HNBR 3. - Example 1: Residual double bond content dropped from 100% to below 1% for the best catalysts by 120-180 min. - Example 1: The process produced HNBR with Mw in the 135-195 kg/mol range and narrowed PDI.

Technical Relevance This is a central HNBR synthesis patent because it directly couples hydrogenation with metathesis to reduce molecular weight and residual unsaturation in one step. It provides explicit catalyst, loading, and kinetic data that are highly relevant to controlled HNBR manufacture.

Patent 6

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low molecular weight HNBR by contacting nitrile rubber with ruthenium- or osmium-based metathesis catalysts in the presence of hydrogen and optionally a co-olefin. The process is aimed at improving hydrogenation efficiency while simultaneously lowering molecular weight and narrowing the molecular weight distribution. - Catalyst amount: 0.041 phr — Example

1, HNBR 1 - Catalyst amount: 0.045 phr — Example 1, HNBR 2 - Catalyst amount: 0.060 phr — Example 1, HNBR 3 - Metal content: 65.6 ppm — Example 1, HNBR 1 - Metal content: 65.6 ppm — Example 1, HNBR 2 - Metal content: 65.6 ppm — Example 1, HNBR 3 - Co-catalyst: NIL — Example 1, HNBR 1 - Co-catalyst: NIL — Example 1, HNBR 2 - Co-catalyst: Triphenylphosphine — Example 1, HNBR 3 - Co-catalyst amount: 1.0 phr — Example 1, HNBR 3 - Reaction time: 0 min — Table 2 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 100% at 0 min — Table 2 - RDB: 2.5% at 60 min for HNBR 1 — Table 2 - RDB: 7.0% at 60 min for HNBR 2 — Table 2 - RDB: 15.4% at 60 min for HNBR 3 — Table 2 - RDB: 0.6% at 120 min for HNBR 1 — Table 2 - RDB: 4.5% at 120 min for HNBR 2 — Table 2 - RDB: 3.1% at 120 min for HNBR 3 — Table 2 - RDB: 0.5% at 180 min for HNBR 1 — Table 2 - RDB: 3.9% at 180 min for HNBR 2 — Table 2 - RDB: 1.3% at 180 min for HNBR 3 — Table 2 - RDB: 0.4% at 240 min for HNBR 1 — Table 2 - RDB: 3.5% at 240 min for HNBR 2 — Table 2 - RDB: 0.7% at 240 min for HNBR 3 — Table 2 - Mw: 113371 g/mol — Table 3, HNBR 1 - Mn: 48352 g/mol — Table 3, HNBR 1 - PDI: 2.34 — Table 3, HNBR 1 - Mooney viscosity: 33.3 — Table 3, HNBR 1 - Mw: 133758 g/mol — Table 3, HNBR 2 - Mn: 54595 g/mol — Table 3, HNBR 2 - PDI: 2.45 — Table 3, HNBR 2 - Mooney viscosity: 41.3 — Table 3, HNBR 2 - Mw: 195304 g/mol — Table 3, HNBR 3 - Mn: 64130 g/mol — Table 3, HNBR 3 - PDI: 3.05 — Table 3, HNBR 3 - Mooney viscosity: 69.3 — Table 3, HNBR 3

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson-type catalysts were compared at similar metal content. - Example 1: HNBR 1 showed the fastest hydrogenation and lowest residual double bond content. - Example 1: Triphenylphosphine co-catalysis was used with the Wilkinson system. - Example 1: The catalyst system reduced molecular weight and Mooney viscosity relative to the industrial norm. - Example 1: The data support catalyst selection as a key lever for low-MW HNBR production.

Technical Relevance This patent is highly relevant because it identifies alternative catalyst systems for simultaneous hydrogenation/metathesis of NBR and quantifies their effect on residual unsaturation and molecular weight. It strengthens the evidence base for ruthenium-based low-MW HNBR synthesis.

Patent 7

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent focuses on hydrogenating NBR to HNBR using ruthenium, palladium, or rhodium compounds while emphasizing catalyst

removal and recovery to preserve product aging properties. The disclosure is process-oriented rather than a single synthetic route, and it highlights the need for controlled catalyst loading, temperature, pressure, and post-hydrogenation purification. - Ruthenium amount: 10 ppm to 200 ppm — claim 1 - Ruthenium amount preferred: 10 ppm to 150 ppm — claim 1 - Ruthenium amount particularly preferred: 10 ppm to 120 ppm — claim 1 - Ruthenium amount very particularly preferred: 10 ppm to 79 ppm — claim 1 - Ruthenium amount most preferred: 42 ppm to 79 ppm — claim 1 - Palladium amount: 20 ppm to <67 ppm — claim 1 - Palladium amount preferred: 44 ppm to <67 ppm — claim 1 - Rhodium amount: 50 ppm to <270 ppm — claim 1 - Rhodium amount preferred: 79 ppm to <270 ppm — claim 1 - Rhodium amount particularly preferred: 50 ppm to 250 ppm — claim 1 - Rhodium amount very particularly preferred: 50 ppm to 170 ppm — claim 1 - Rhodium amount most preferred: 79 ppm to 170 ppm — claim 1 - Hydrogenation temperature: 60 to 200°C — claim 1 - Hydrogenation pressure: 700000 to 15000000 Pa — claim 1 - Hydrogenation time: 1 to 24 hours — claim 1 - HNBR composition: 40% to 90% conjugated diene units — claim 1 - HNBR composition: 10% to 60% α,β -unsaturated nitrile units — claim 1 - HNBR ageing criterion: gel content increase less than 10% after 4 days at 140°C — description - HNBR ageing criterion: Mooney viscosity increase less than 40% after 4 days at 140°C — description

Relevant Experimental Evidence - No parser-detected examples were provided; the patent is supported mainly by claims and detailed description. - The disclosure emphasizes catalyst residue removal as essential for color and aging stability. - The patent defines broad Ru/Pd/Rh loading windows for direct HNBR hydrogenation. - The patent links low catalyst residue to improved gel content and Mooney aging behavior.

Technical Relevance This patent is important because it addresses a practical bottleneck in HNBR manufacture: catalyst residue removal and its impact on aging, color, and gel content. It provides explicit catalyst loading and hydrogenation windows that are useful for designing robust HNBR production processes.

Patent 8

- Patent Number: US10731020B2
- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a direct HNBR synthesis patent; it discloses rubber compositions for vulcanization molding that may include hydrogenated nitrile rubber among many possible base rubbers. The key process concept is the use of hollow particles with controlled expansion ratio in a rubber matrix to achieve lightweight molded products with good dimensional stability and appearance. - Base rubber Mooney viscosity

ML(1+4) at 100°C: 97 — Comparative Example 1 - Kneading temperature: 60°C — Comparative Example 1 - Kneading time: about 10 minutes — Comparative Example 1 - Vulcanization temperature: 140°C — Comparative Example 1 - Vulcanization time: 30 minutes — Comparative Example 1 - Molding pressure: 15 MPa — Comparative Example 1 - Further expansion ratio: 50.0% — Example 1 - Base rubber Mooney viscosity ML(1+4) at 100°C: 54 — Example 1 - Process oil: 20 parts by weight — Example 1 - Vulcanization accelerator: 3.5 parts by weight — Example 1 - Sulfur: 0.4 parts by weight — Example 1 - Kneading temperature: 60°C — Example 1 - Kneading time: about 10 minutes — Example 1 - Theoretical true specific gravity: 0.96 — Example 1 - Actual true specific gravity: 0.97 — Example 1 - Breakage ratio: 3.71% — Example 1 - Vulcanization molding temperature: 140°C — Example 1 - Further expansion ratio: 22.2% — Table 4 / Example 2-5 - Base rubber Mooney viscosity ML(1+4) at 100°C: 42 — Table 4 - Process oil: 50 parts by weight — Table 4 - Vulcanization accelerator: 4 parts by weight — Table 4 - Sulfur: 0.6 parts by weight — Table 4 - Further expansion ratio: 72.4% — Table 4 - Base rubber Mooney viscosity ML(1+4) at 100°C: 73 — Table 4 - Vulcanization molding temperature: 150°C — Example 3 - Vulcanization molding temperature: 170°C — Example 5 - Vulcanization molding temperature: 180°C — Example 6 - Injection molding cylinder temperature: 180°C — Example 6 - Injection molding die temperature: 60°C — Example 6 - Injection pressure: 15 MPa — Example 6 - Mean particle size: 129.8 μm — Example 11

Relevant Experimental Evidence - Example 1: Hollow particles with 50% further expansion ratio gave low breakage and excellent lightweight performance. - Example 3: A 22.2% further expansion ratio and 150°C vulcanization produced a stable, lightweight molded product. - Example 6: Injection molding at 180°C with 72.4% further expansion ratio still gave acceptable dimensional stability. - Example 11: Large-particle hollow fillers produced acceptable products with only minor surface unevenness. - Comparative examples showed that too low or too high expansion ratios caused breakage or re-expansion defects.

Technical Relevance This patent is only indirectly relevant to HNBR synthesis, but it is useful for downstream processing of HNBR-containing rubber compositions. It provides compounding and molding conditions that can preserve lightweight filler integrity in HNBR-based formulations.

Patent 9

- Patent Number: US10641418B2
- Patent Title: Chemical and gas-resistant, high-pressure, large-bore bonded flexible rubber ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a bonded flexible pipe in which a peroxide-vulcanized HNBR-based rubber compound is directly bonded to a PVDF

liner. The HNBR is used as a chemically resistant layer in a high-pressure hose structure, and the key process variables are scorch time, peroxide selection, and vulcanization temperature. - Internal diameter: 4 inch (102 mm) — Example 1 - Operating pressure: 69 MPa — Example 1 - PVDF liner thickness: 6 mm — Example 1 - HNBR rubber compound layer thickness: 1 mm — Example 1 - Mooney 05, 150°C scorch time: 12.0 minutes — Example 1 - Trigonox 145 amount: 8 phr — Example 1 - Trigonox 145 active agent content: 45% — Example 1 - Trigonox 145 half-life at 154°C: 1 hour — Example 1 - Acrylonitrile content: 30% to 40% — Example 1 - Residual double bond content: below 2% — Example 1 - Carbon black: 70 phr — Example 1 - Triallyl isocyanate coagent: 2.5 phr — Example 1 - Other additives: 18.7 phr — Example 1 - Vulcanization temperature: 175°C — Example 1 - Internal diameter: 4 inch (102 mm) — Example 2 - Mooney 05, 150°C scorch time: 5.6 minutes — Example 2 - Perkadox 1440 amount: 8 mass fractions — Example 2 - Perkadox 1440 active substance: 40% — Example 2 - Perkadox 1440 half-life at 140°C: 1 hour — Example 2 - Vulcanization temperature: 155°C — Example 2 - Internal diameter: 3 inch (76 mm) — Example 3 - Operating pressure: 103.5 MPa — Example 3 - PVDF liner melting point: 166°C to 170°C — Example 3 - HNBR rubber compound layer thickness: 1 mm — Example 3 - Mooney 05, 150°C scorch time: 14.0 minutes — Example 3 - Luperox F40P-SP2 amount: 7 phr — Example 3 - Vulcanization temperature: 160°C — Example 3

Relevant Experimental Evidence - Example 1: A 4-inch pipe with 69 MPa operating pressure achieved cohesive failure and adhesion above 10 N/mm. - Example 2: Short scorch time and lower vulcanization temperature led to adhesive failure and only 3.1 N/mm adhesion. - Example 3: A 3-inch pipe at 103.5 MPa achieved cohesive failure and adhesion above 15 N/mm. - Example 3: Scorch-protected peroxide and 160°C vulcanization improved bonding to PVDF. - Example 3: The pipe was described as resistant to zinc bromide and strong acids in oil-well service.

Technical Relevance This patent is relevant because it shows how HNBR is formulated and cured for demanding bonded pipe applications, where low residual unsaturation and controlled scorch behavior are critical. It provides practical peroxide and cure-temperature guidance for HNBR-based compounds.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- The metathesis-assisted HNBR patents (US7919563B2, US7772328B2, EP2473280B1, US9150669B2) all rely on ruthenium/osmium catalysts and elevated temperature, with 80°C explicitly shown in the low-MW examples and hydrogen/co-olefin assistance used to drive chain scission.
- US7919563B2 and US7772328B2 disclose very similar process windows: 15 wt.-% or 6 wt.-% cement, 6.9×10^5 Pa ethylene, 150 g 1-hexene in one example, and catalyst loadings of 0.05-0.25 phr; both report Mw reductions to the 50-60 kg/mol range and PDI near 2.0 in the best case.

- EP2473280B1 and US9150669B2 compare catalyst systems directly; EP2473280B1 shows Grubbs III outperforming Grubbs II and Wilkinson's catalyst at equal metal content, while US9150669B2 shows Hoveyda/Arlt-type catalysts outperforming Wilkinson's catalyst in hydrogenation rate and molecular-weight reduction.
- EP2473280B1 and US9150669B2 both emphasize simultaneous hydrogenation/metathesis, but EP2473280B1 reports lower residual double bond contents more rapidly with Grubbs III, whereas US9150669B2 highlights the benefit of catalyst/co-catalyst selection and reports PDI values down to 2.34.
- EP3827029A1 differs from the metathesis patents by focusing on conventional hydrogenation with Ru/Pd/Rh loadings in the ppm range and broad conditions of 60-200°C, 700000-15000000 Pa, and 1-24 h, with catalyst removal/recovery as a central process issue.
- The application patents (US20130303700A1, US10759885B2, US10731020B2, US10641418B2) do not primarily synthesize HNBR but show how partially hydrogenated or hydrogenated rubber is compounded: metathesis blending with silicone, hydrogenated branched-diene precursor design, hollow-particle lightweight molding, and peroxide-vulcanized bonded pipe construction.
- US10641418B2 is the most application-specific HNBR disclosure, requiring Mooney 05, 150°C scorch time of at least 10 minutes and vulcanization at at least 160°C to achieve strong PVDF adhesion, whereas US10731020B2 uses HNBR only as one possible base rubber in a lightweight molding composition.
- Across the synthesis patents, low residual unsaturation is consistently linked to performance: EP2473280B1 and US9150669B2 target low RDB, while US10641418B2 specifies residual double bond content below 2% for the HNBR compound used in the pipe.

4. CONCLUSION

The evidence supports two main HNBR manufacturing strategies: direct hydrogenation with careful catalyst management, and simultaneous hydrogenation/metathesis to lower molecular weight and Mooney viscosity. For synthesis, the strongest evidence favors ruthenium-based homogeneous catalysts, hydrogenation temperatures around 80°C for metathesis-assisted routes, and catalyst loadings in the low phr or ppm range, with co-olefins such as ethylene or 1-hexene used to promote controlled chain scission. For conventional hydrogenation, broad windows of 60-200°C and 700000-15000000 Pa are disclosed, but catalyst residue removal is critical to preserve aging and color stability. For downstream HNBR compounding, peroxide systems with scorch times of at least 10 minutes and cure temperatures of 160-175°C appear beneficial in demanding bonded structures. Overall, the best-supported synthesis parameters are low catalyst loading, controlled hydrogen pressure, moderate temperature, and post-reaction purification or catalyst recovery to achieve low residual unsaturation, low Mooney viscosity, and good long-term performance.

5. REFERENCES

- US20130303700A1 | Blends of partially hydrogenated nitrile rubber and silicone rubber, ... | Not disclosed | US | 2013 | <https://patents.google.com/patent/US20130303700A1/en>
- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- US10759885B2 | Hydrogenated branched conjugated diene copolymer, rubber composition and ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10759885B2/en>
- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- US10731020B2 | Rubber composition for vulcanization molding, process for manufacturing the ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731020B2/en>
- US10641418B2 | Chemical and gas-resistant, high-pressure, large-bore bonded flexible rubber ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10641418B2/en>